

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

***CO1:** Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05

Total Marks:100

Annexure A

Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

*I G SAI TEJA (192472137) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes

P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes
P20	29	92	112	22	No

3. Retail Sales Dataset

Question 3

A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

4. Banking Loan Approval Dataset

Question 4

A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected

C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

5. Agriculture Crop Yield Dataset

Question 5

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis
Insight Generation and Reporting	15	Presents strong insights, conclusions, and recommendations based on data evidence	Presents clear findings and conclusions with reasonable support	Provides basic findings but lacks depth or proper justification	Findings are unclear, unsupported, or not related to the data

CO1-ASSESSMENT TOOL -2 Concept Map

QUESTIONS : 5

MARKS : 50

Q No	Dataset Understanding (20%)	Data Cleaning (25%)	Data Analysis and Statistical Summary (20%)	Data Visualization (20%)	Insight Generation and Reporting (15%)	Total Score (100%)
1	/2.5	/2.5	/2.0	/2.0	/1.0	/10
2	/2.5	/2.5	/2.0	/2.0	/1.0	/10
3	/2.5	/2.5	/2.0	/2.0	/1.0	/10
4	/2.5	/2.5	/2.0	/2.0	/1.0	/10
5	/2.5	/2.5	/2.0	/2.0	/1.0	/10
Overall Rubrics Value	/12.5	/12.5	/10	/10	/10	/50
	/25	/25	/20	/20	/10	/100

```

# ===== STEP 1: Import Libraries =====
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# ===== STEP 2: Create Dataset =====
data = {
    'Student_ID':
    ['S01', 'S02', 'S03', 'S04', 'S05', 'S06', 'S07', 'S08', 'S09', 'S10',
    'S11', 'S12', 'S13', 'S14', 'S15', 'S16', 'S17', 'S18', 'S19', 'S20'],
    'Attendance':
    [92, 55, 78, 60, 88, 47, 95, 72, 50, 83, 66, 90, 58, 76, 62, 85, 45, 80, 68, 53],
    'Study_Hours': [5, 2, 4, 3, 5, 1, 6, 3, 2, 4, 3, 5, 2, 4, 3, 5, 1, 4, 3, 2],
    'Internal_Marks':
    [45, 24, 38, 28, 42, 20, 48, 35, 22, 40, 30, 44, 25, 36, 29, 41, 18, 39, 32, 23],
    'Result':
    ['Pass', 'Fail', 'Pass', 'Fail', 'Pass', 'Fail', 'Pass', 'Pass', 'Fail', 'Pass',
    ,
    'Fail', 'Pass', 'Fail', 'Pass', 'Fail', 'Pass', 'Fail', 'Pass', 'Pass', 'Fail']
}

df = pd.DataFrame(data)
print("===== DATASET =====")
print(df)

# ===== STEP 3: Basic Data Overview =====
print("\n===== SHAPE =====")
print(df.shape)

print("\n===== DATA TYPES =====")
print(df.dtypes)

print("\n===== MISSING VALUES =====")
print(df.isnull().sum())

print("\n===== SUMMARY STATISTICS =====")
print(df.describe())

# ===== STEP 4: Convert Result to Numeric =====
df['Result_Num'] = df['Result'].map({'Pass': 1, 'Fail': 0})
print("\n===== RESULT ENCODED =====")
print(df[['Student_ID', 'Result', 'Result_Num']])

# ===== STEP 5: Correlation Analysis =====
correlation = df[['Attendance', 'Study_Hours', 'Internal_Marks',
'Result_Num']].corr()
print("\n===== CORRELATION MATRIX =====")

```

```

print(correlation)

plt.figure(figsize=(6,4))
sns.heatmap(correlation, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# ===== STEP 6: Group-wise Analysis (Pass vs Fail) =====
print("\n===== AVERAGE VALUES: PASS vs FAIL =====")
print(df.groupby('Result')[['Attendance', 'Study_Hours',
'Internal_Marks']].mean())

# ===== STEP 7: Visualizations =====

# Attendance vs Result
plt.figure(figsize=(6,4))
sns.boxplot(x='Result', y='Attendance', data=df)
plt.title("Attendance vs Result")
plt.show()

# Study Hours vs Result
plt.figure(figsize=(6,4))
sns.boxplot(x='Result', y='Study_Hours', data=df)
plt.title("Study Hours vs Result")
plt.show()

# Internal Marks vs Result
plt.figure(figsize=(6,4))
sns.boxplot(x='Result', y='Internal_Marks', data=df)
plt.title("Internal Marks vs Result")
plt.show()

# Scatter plot: Attendance vs Internal Marks colored by Result
plt.figure(figsize=(6,4))
sns.scatterplot(x='Attendance', y='Internal_Marks', hue='Result',
data=df, s=100)
plt.title("Attendance vs Internal Marks by Result")
plt.show()

# Histogram of Attendance
plt.figure(figsize=(6,4))
sns.histplot(df['Attendance'], bins=8, kde=True)
plt.title("Distribution of Attendance")
plt.show()

# Histogram of Study Hours
plt.figure(figsize=(6,4))
sns.histplot(df['Study_Hours'], bins=6, kde=True)
plt.title("Distribution of Study Hours")
plt.show()

```



```

# ===== STEP 8: Insights =====
print("\n===== KEY INSIGHTS =====")
pass_avg = df[df['Result']=='Pass']
[['Attendance', 'Study_Hours', 'Internal_Marks']].mean()
fail_avg = df[df['Result']=='Fail']
[['Attendance', 'Study_Hours', 'Internal_Marks']].mean()

print(f"Average Attendance - Pass: {pass_avg['Attendance']:.2f}, Fail: {fail_avg['Attendance']:.2f}")
print(f"Average Study Hours - Pass: {pass_avg['Study_Hours']:.2f}, Fail: {fail_avg['Study_Hours']:.2f}")
print(f"Average Internal Marks - Pass: {pass_avg['Internal_Marks']:.2f}, Fail: {fail_avg['Internal_Marks']:.2f}")

print("\nConclusion: Students with higher attendance, more study hours, and higher internal marks are more likely to Pass. Attendance and Internal Marks show the strongest relationship with the final result, based on the correlation analysis and group averages.")

```

```

===== DATASET =====

```

	Student_ID	Attendance	Study_Hours	Internal_Marks	Result
0	S01	92	5	45	Pass
1	S02	55	2	24	Fail
2	S03	78	4	38	Pass
3	S04	60	3	28	Fail
4	S05	88	5	42	Pass
5	S06	47	1	20	Fail
6	S07	95	6	48	Pass
7	S08	72	3	35	Pass
8	S09	50	2	22	Fail
9	S10	83	4	40	Pass
10	S11	66	3	30	Fail
11	S12	90	5	44	Pass
12	S13	58	2	25	Fail
13	S14	76	4	36	Pass
14	S15	62	3	29	Fail
15	S16	85	5	41	Pass
16	S17	45	1	18	Fail
17	S18	80	4	39	Pass
18	S19	68	3	32	Pass
19	S20	53	2	23	Fail

```

===== SHAPE =====
(20, 5)

```

```

===== DATA TYPES =====

```

```
Student_ID      object
Attendance      int64
Study_Hours     int64
Internal_Marks  int64
Result          object
dtype: object
```

===== MISSING VALUES =====

```
Student_ID      0
Attendance      0
Study_Hours     0
Internal_Marks  0
Result          0
dtype: int64
```

===== SUMMARY STATISTICS =====

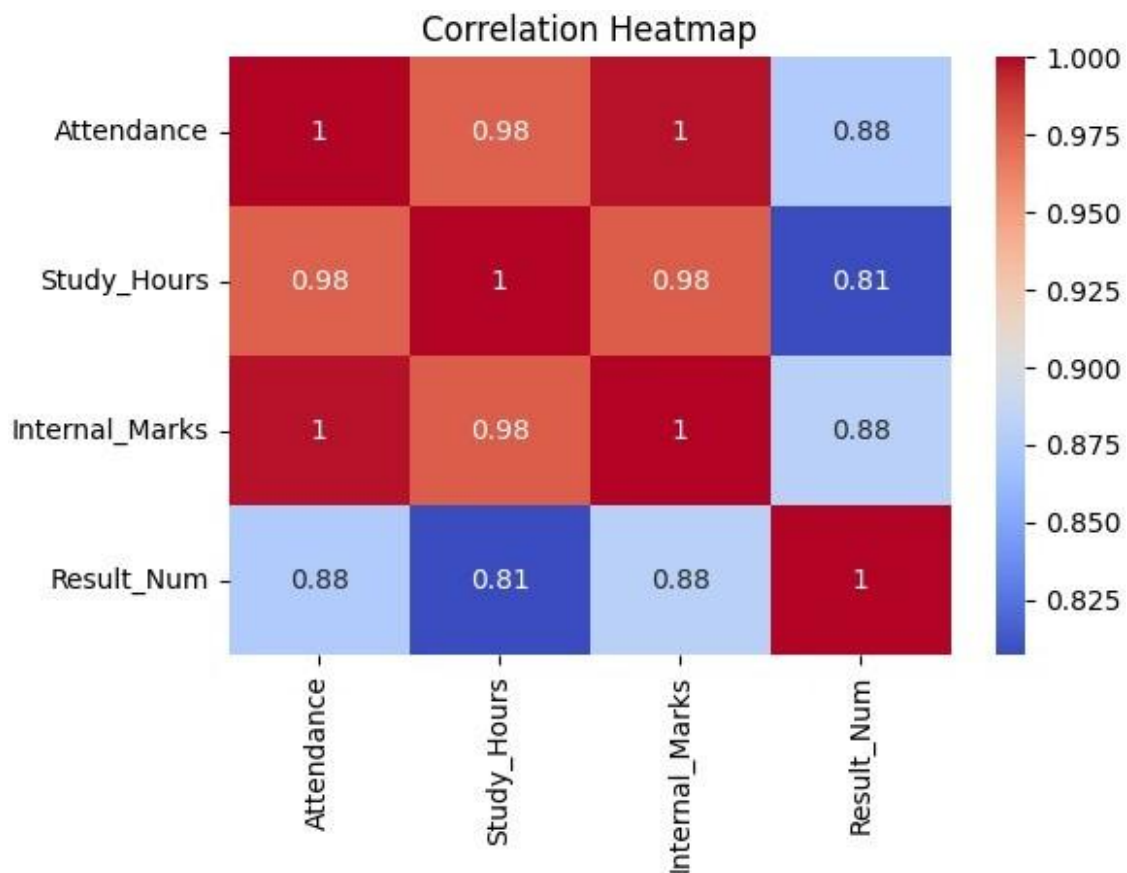
	Attendance	Study_Hours	Internal_Marks
count	20.000000	20.000000	20.000000
mean	70.150000	3.350000	32.950000
std	15.938286	1.424411	9.087151
min	45.000000	1.000000	18.000000
25%	57.250000	2.000000	24.750000
50%	70.000000	3.000000	33.500000
75%	83.500000	4.250000	40.250000
max	95.000000	6.000000	48.000000

===== RESULT ENCODED =====

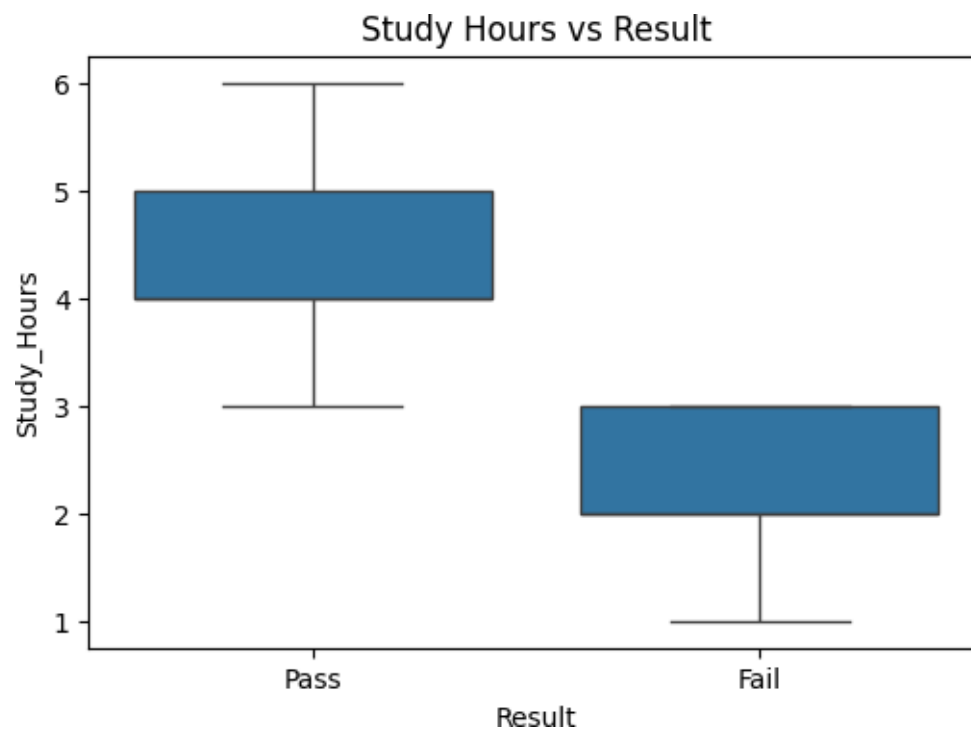
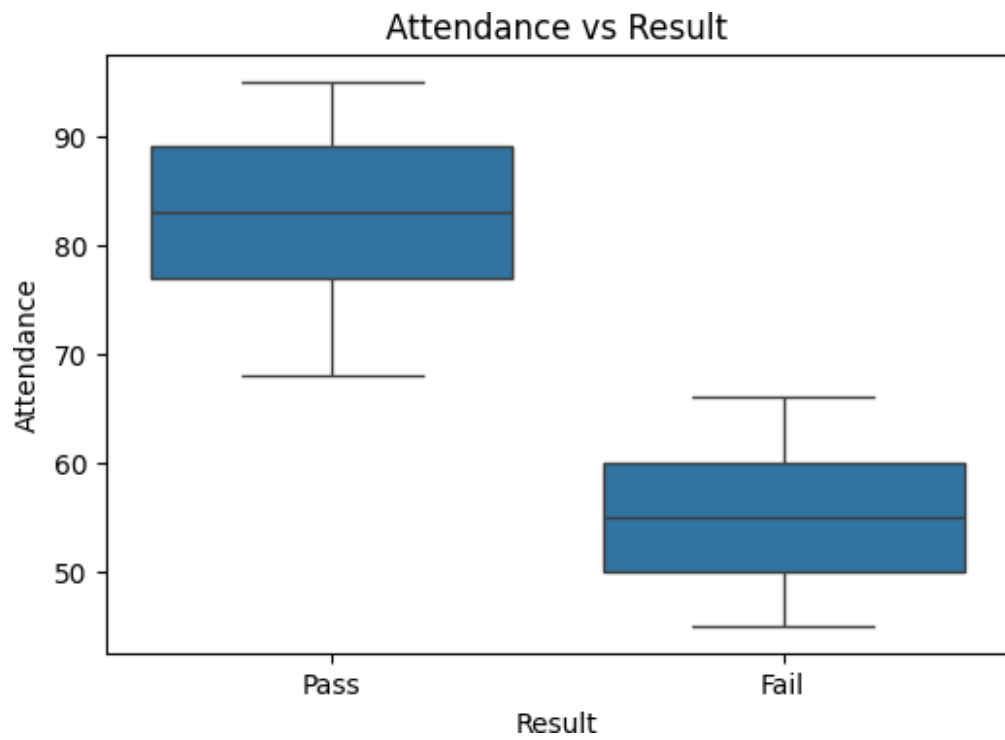
	Student ID	Result	Result_Num
0	S01	Pass	1
1	S02	Fail	0
2	S03	Pass	1
3	S04	Fail	0
4	S05	Pass	1
5	S06	Fail	0
6	S07	Pass	1
7	S08	Pass	1
8	S09	Fail	0
9	S10	Pass	1
10	S11	Fail	0
11	S12	Pass	1
12	S13	Fail	0
13	S14	Pass	1
14	S15	Fail	0
15	S16	Pass	1
16	S17	Fail	0
17	S18	Pass	1
18	S19	Pass	1
19	S20	Fail	0

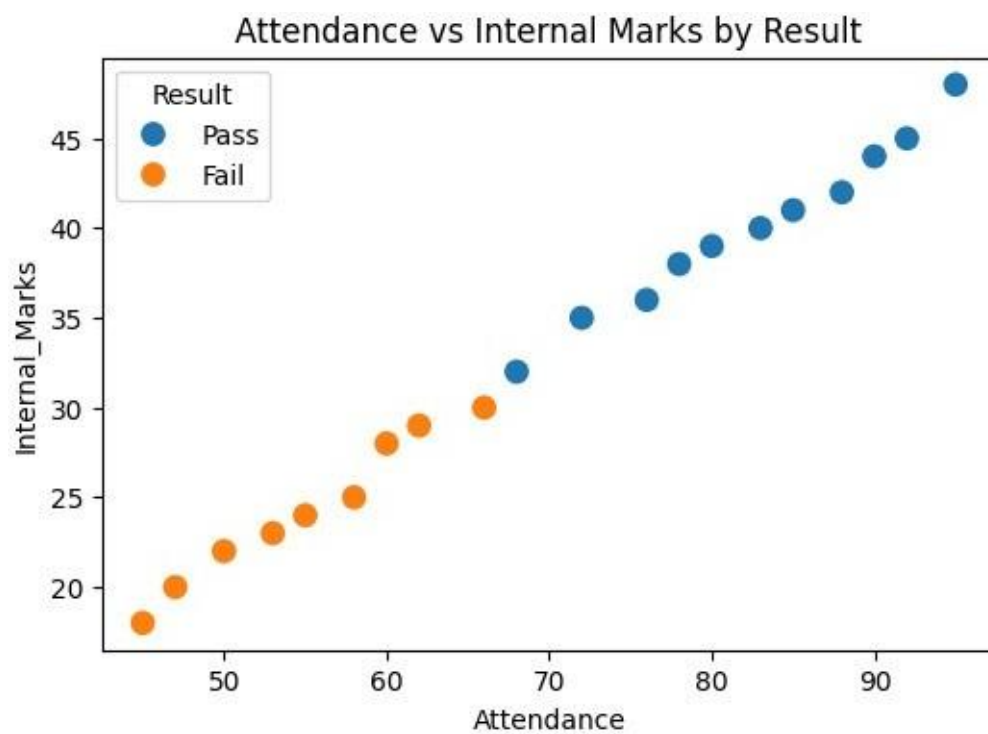
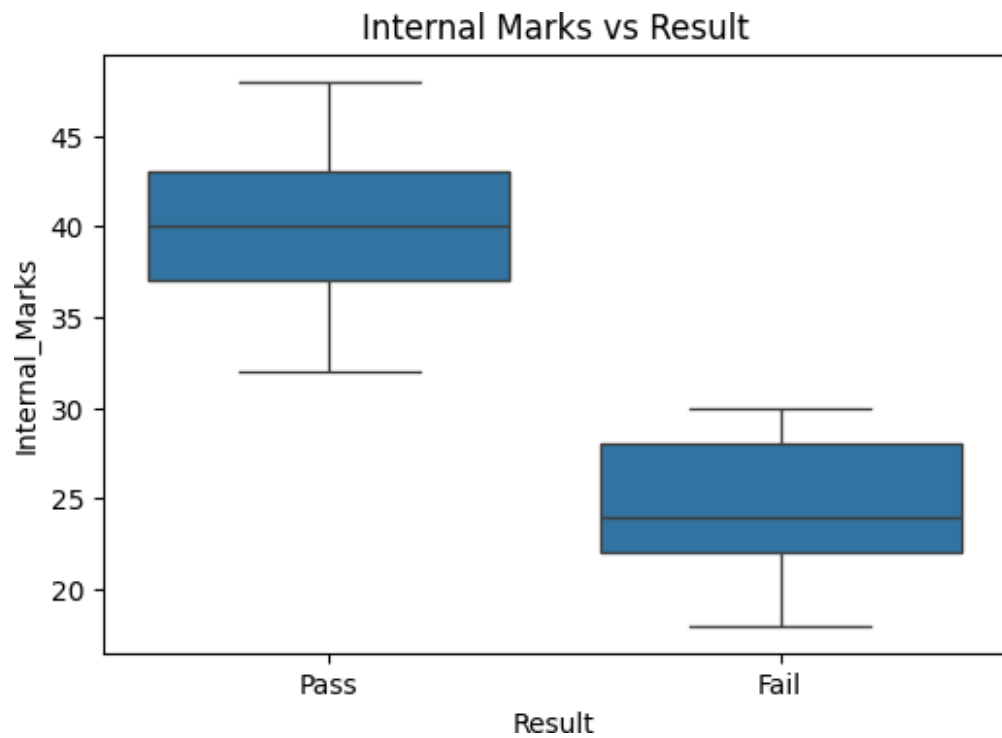
===== CORRELATION MATRIX =====

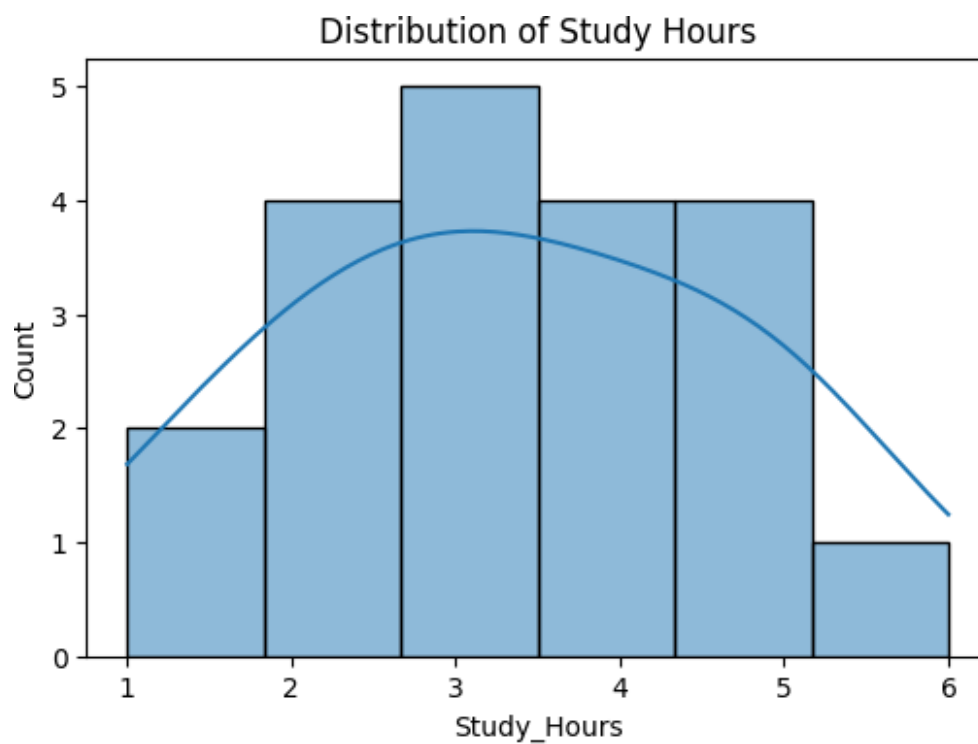
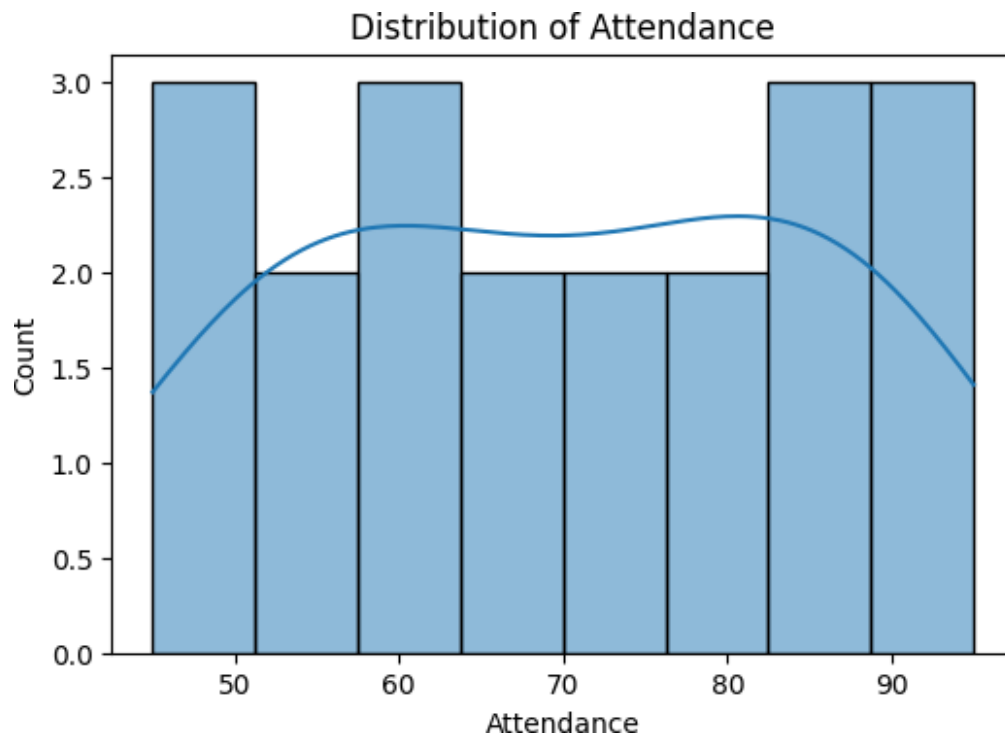
	Attendance	Study_Hours	Internal_Marks	Result_Num
Attendance	1.000000	0.975888	0.997569	0.875664
Study_Hours	0.975888	1.000000	0.977299	0.807161
Internal_Marks	0.997569	0.977299	1.000000	0.879984
Result_Num	0.875664	0.807161	0.879984	1.000000



===== AVERAGE VALUES: PASS vs FAIL =====			
	Attendance	Study_Hours	Internal_Marks
Result			
Fail	55.111111	2.111111	24.333333
Pass	82.454545	4.363636	40.000000







===== KEY INSIGHTS =====

Average Attendance - Pass: 82.45, Fail: 55.11

Average Study Hours - Pass: 4.36, Fail: 2.11
Average Internal Marks - Pass: 40.00, Fail: 24.33

Conclusion: Students with higher attendance, more study hours, and higher internal marks are more likely to Pass. Attendance and Internal Marks show the strongest relationship with the final result, based on the correlation analysis and group averages.

```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
    "Patient_ID": ["P01", "P02", "P03", "P04", "P05", "P06", "P07",
                  "P08", "P09", "P10",
                  "P11", "P12", "P13", "P14", "P15", "P16", "P17",
                  "P18", "P19", "P20"],
    "Age": [45, 32, 55, 28, 60, 40, 52, 35, 48, 30,
            62, 38, 50, 27, 58, 42, 46, 34, 53, 29],
    "Sugar_Level": [145, 98, 180, 90, 200, 120, 165, 105, 155, 95,
                   210, 115, 170, 88, 190, 130, 150, 100, 175, 92],
    "BP": [130, 118, 145, 110, 150, 125, 140, 120, 135, 115,
           155, 122, 142, 108, 148, 128, 132, 116, 144, 112],
    "BMI": [28, 23, 31, 22, 34, 26, 30, 24, 29, 23,
            35, 25, 30, 21, 33, 27, 28, 24, 31, 22],
    "Disease_Status": ["Yes", "No", "Yes", "No", "Yes", "No", "Yes",
                      "No", "Yes", "No",
                      "Yes", "No", "Yes", "No", "Yes", "No", "Yes",
                      "No", "Yes", "No"]
}

df = pd.DataFrame(data)

# 1. Basic Info and Stats
print("Healthcare Analytics Dataset Summary")
print(df.info())
print("\nMissing Values:\n", df.isnull().sum())
print("\nStatistical Measures:\n", df.describe())

# 2. Charts
plt.figure(figsize=(12, 8))

# Sugar Level Distribution
plt.subplot(2, 2, 1)
sns.histplot(df["Sugar_Level"], bins=5, kde=True, color="skyblue")
plt.title("Sugar Level Distribution")
```

```

# Disease Status Count
plt.subplot(2, 2, 2)
sns.countplot(x="Disease_Status", data=df)
plt.title("Disease Status Count")

# Sugar Level vs BMI
plt.subplot(2, 2, 3)
sns.scatterplot(x="Sugar_Level", y="BMI", hue="Disease_Status",
data=df)
plt.title("Sugar Level vs BMI")

# Box Plot Sugar by Status
plt.subplot(2, 2, 4)
sns.boxplot(x="Disease_Status", y="Sugar_Level", data=df)
plt.title("Sugar Level by Disease Status")

plt.tight_layout()
plt.show()

# 3. Observations and Conclusion
print("\nObservations")
print("1. High sugar level increases disease risk.")
print("2. Patients with higher BMI are more likely to have disease.")
print("3. Sugar Level is the strongest indicator of disease in this
dataset.")
print("\nConclusion: Sugar Level, BMI, and BP are key factors
associated with disease status.")

```

```

Healthcare Analytics Dataset Summary
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Patient_ID      20 non-null    object
1   Age             20 non-null    int64
2   Sugar_Level     20 non-null    int64
3   BP              20 non-null    int64
4   BMI             20 non-null    int64
5   Disease_Status  20 non-null    object
dtypes: int64(4), object(2)
memory usage: 1.1+ KB
None

```

```

Missing Values:
Patient_ID      0
Age             0
Sugar_Level     0
BP              0

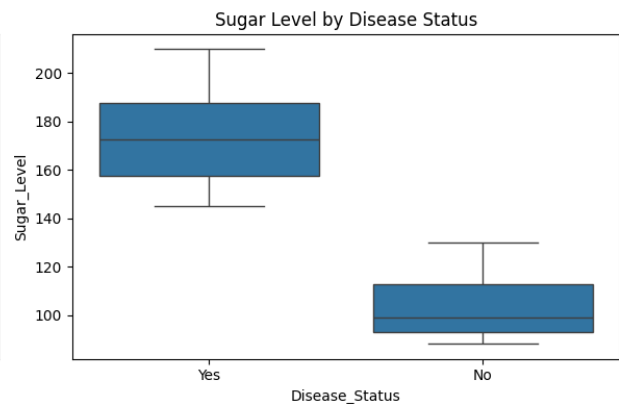
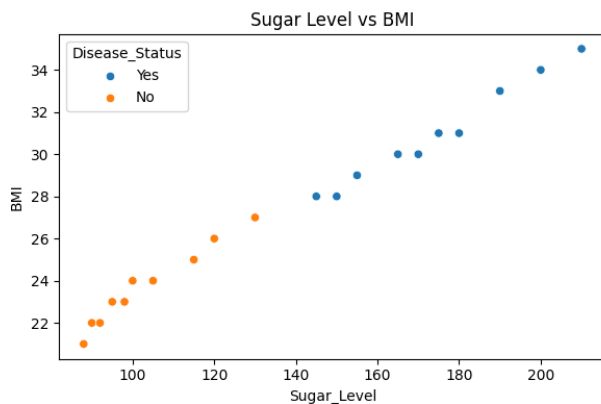
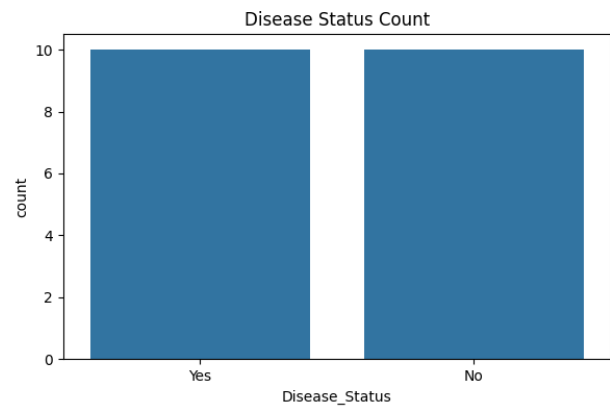
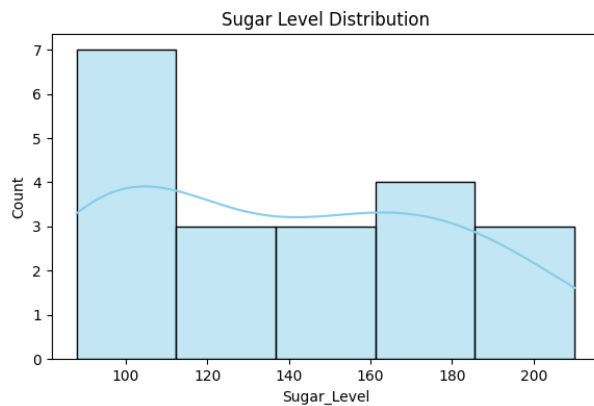
```



```
BMI      0
Disease Status  0
dtype: int64
```

Statistical Measures:

	Age	Sugar Level	BP	BMI
count	20.000000	20.000000	20.000000	20.000000
mean	43.200000	138.650000	129.750000	27.300000
std	11.316499	40.33449	14.523574	4.256265
min	27.000000	88.000000	108.000000	21.000000
25%	33.500000	99.500000	117.500000	23.750000
50%	43.500000	137.500000	129.000000	27.500000
75%	52.250000	171.250000	142.500000	30.250000
max	62.000000	210.000000	155.000000	35.000000



Observations

1. High sugar level increases disease risk.
2. Patients with higher BMI are more likely to have disease.
3. Sugar Level is the strongest indicator of disease in this dataset.

Conclusion: Sugar Level, BMI, and BP are key factors associated with disease status.

```

# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
    "Product_ID":
    ["PR01", "PR02", "PR03", "PR04", "PR05", "PR06", "PR07", "PR08", "PR09", "PR10"
    ,
    "PR11", "PR12", "PR13", "PR14", "PR15", "PR16", "PR17", "PR18", "PR19", "PR20"]
    ,
    "Category": ["Grocery", "Snacks", "Dairy", "Fruits", "Vegetables",
    "Grocery", "Snacks", "Dairy", "Fruits", "Vegetables",
    "Grocery", "Snacks", "Dairy", "Fruits", "Vegetables",
    "Grocery", "Snacks", "Dairy", "Fruits", "Vegetables"],
    "Price": [50, 30, 45, 60, 40, 80, 25, 55, 70, 35,
    90, 20, 65, 75, 45, 100, 35, 50, 80, 30],
    "Quantity_Sold": [20, 35, 18, 25, 30, 15, 40, 16, 22, 28,
    12, 45, 14, 20, 26, 10, 32, 19, 18, 35],
    "Discount": [5, 3, 2, 5, 4, 6, 3, 2, 5, 4,
    6, 2, 3, 5, 4, 7, 3, 2, 6, 3],
    "Revenue": [1000, 1050, 810, 1500, 1200, 1200, 1000, 880, 1540, 980,
    1080, 900, 910, 1500, 1170, 1000, 1120, 950, 1440, 1050]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display Dataset
print("Retail Sales Dataset")
print(df.head())

# 1. Dataset Dimensions
print("\nRows and Columns:", df.shape)

# 2. Data Integrity
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Values:", df.duplicated().sum())

# 3. Statistical Measures
print("\nStatistical Summary (Mean):")
print(df.mean(numeric_only=True))

# ----- Visualizations -----
plt.figure(figsize=(14, 10))

# 1. Revenue Distribution

```

```

plt.subplot(2, 2, 1)
sns.histplot(df["Revenue"], bins=5, kde=True, color='green')
plt.title("Revenue Distribution")

# 2. Revenue by Category
plt.subplot(2, 2, 2)
category_revenue = df.groupby("Category")["Revenue"].sum()
category_revenue.plot(kind="bar", color="skyblue")
plt.title("Total Revenue by Category")
plt.ylabel("Revenue")

# 3. Price vs Revenue Scatter Plot
plt.subplot(2, 2, 3)
sns.scatterplot(x="Price", y="Revenue", hue="Category", data=df,
s=100)
plt.title("Price vs Revenue")

# 4. Revenue Box Plot by Category
plt.subplot(2, 2, 4)
sns.boxplot(x="Category", y="Revenue", data=df)
plt.title("Revenue Spread by Category")

plt.tight_layout()
plt.show()

# 4. Insights and Conclusions
print("\nImportant Patterns & Observations")
print("1. Fruits category generated the highest total revenue.")
print("2. Snacks recorded the highest quantity sold volume.")
print("3. Revenue shows a positive correlation with product price.")
print("4. Discounts have a secondary impact compared to base price and volume.")

print("\nConclusion")
print("Price and Quantity Sold are the primary drivers of revenue. The Fruits category is the most profitable, while Snacks drive the highest inventory turnover.")

```

Retail Sales Dataset

	Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
0	PR01	Grocery	50	20	5	1000
1	PR02	Snacks	30	35	3	1050
2	PR03	Dairy	45	18	2	810
3	PR04	Fruits	60	25	5	1500
4	PR05	Vegetables	40	30	4	1200

Rows and Columns: (20, 6)

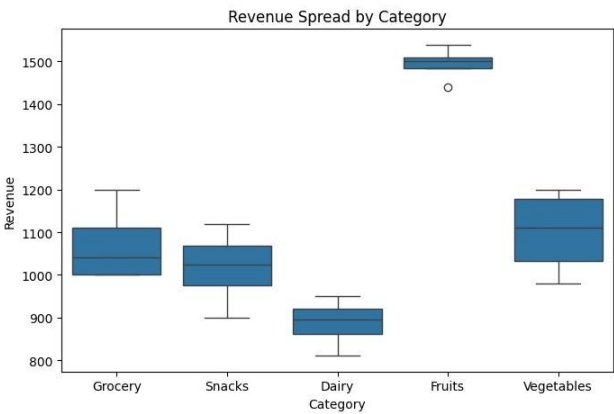
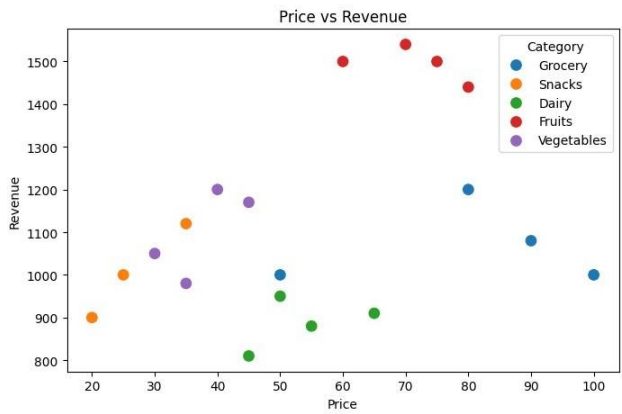
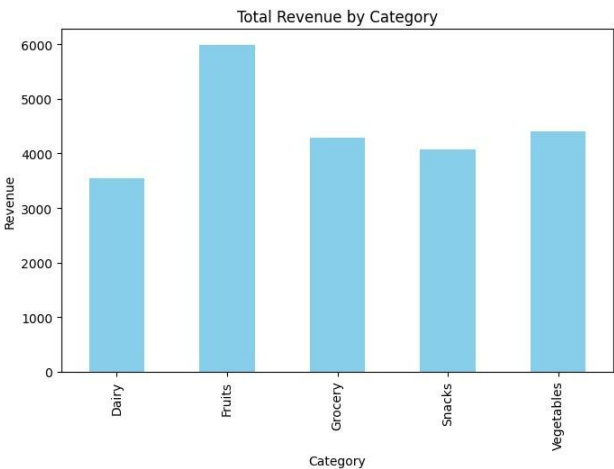
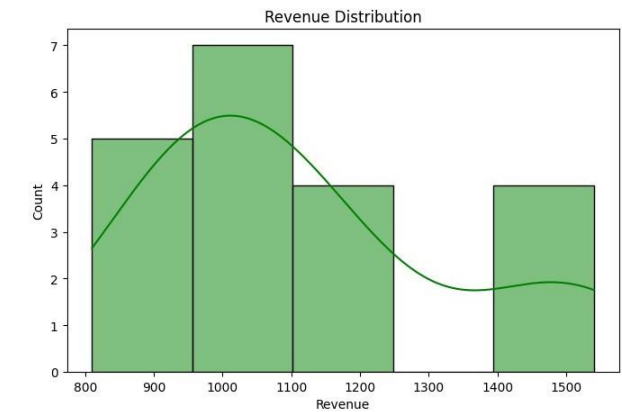
Missing Values:

Product_ID 0

```
Category      0
Price         0
Quantity_Sold 0
Discount      0
Revenue       0
dtype: int64
```

Duplicate Values: 0

```
Statistical Summary (Mean):
Price          54.0
Quantity_Sold  24.0
Discount       4.0
Revenue       1114.0
dtype: float64
```



Important Patterns & Observations

1. Fruits category generated the highest total revenue.
2. Snacks recorded the highest quantity sold volume.
3. Revenue shows a positive correlation with product price.
4. Discounts have a secondary impact compared to base price and

volume.

Conclusion

Price and Quantity Sold are the primary drivers of revenue. The Fruits category is the most profitable, while Snacks drive the highest inventory turnover.

```

# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
    "Customer_ID":
    ["C01", "C02", "C03", "C04", "C05", "C06", "C07", "C08", "C09", "C10",
    "C11", "C12", "C13", "C14", "C15", "C16", "C17", "C18", "C19", "C20"],
    "Income":
    [45000, 25000, 60000, 22000, 50000, 30000, 70000, 28000, 55000, 24000,
    65000, 35000, 48000, 26000, 75000, 32000, 52000, 27000, 68000, 23000],
    "Credit_Score": [720, 580, 750, 560, 710, 600, 780, 590, 730, 570,
    760, 620, 700, 585, 790, 610, 725, 575, 770, 555],
    "Loan_Amount":
    [200000, 150000, 300000, 120000, 250000, 180000, 350000, 160000, 270000, 130000,
    ,
    320000, 200000, 240000, 140000, 400000, 190000, 260000, 150000, 330000, 125000]
    ,
    "Employment_Type": ["Salaried", "Self-
    employed", "Salaried", "Unemployed", "Salaried",
    "Self-employed", "Salaried", "Self-
    employed", "Salaried", "Unemployed",
    "Salaried", "Self-
    employed", "Salaried", "Unemployed", "Salaried",
    "Self-employed", "Salaried", "Self-
    employed", "Salaried", "Unemployed"],
    "Loan_Status":
    ["Approved", "Rejected", "Approved", "Rejected", "Approved",
    "Rejected", "Approved", "Rejected", "Approved", "Rejected",
    "Approved", "Rejected", "Approved", "Rejected", "Approved",
    "Rejected", "Approved", "Rejected", "Approved", "Rejected"]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display Dataset
print("Banking Loan Approval Dataset")
print(df)

# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

```

```

# 2. Data Types
print("\nData Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# ----- Charts -----

# 1. Histogram
plt.figure(figsize=(6,4))
sns.histplot(df["Credit_Score"], bins=5, kde=True)
plt.title("Credit Score Distribution")
plt.show()

# 2. Bar Chart
loan_count = df["Loan_Status"].value_counts()

plt.figure(figsize=(5,4))
loan_count.plot(kind="bar", color=["green", "red"])
plt.title("Loan Approval Status")
plt.xlabel("Loan Status")
plt.ylabel("Count")
plt.show()

# 3. Scatter Plot
plt.figure(figsize=(6,4))
sns.scatterplot(x="Income", y="Credit_Score", hue="Loan_Status",
data=df)

```

```

plt.title("Income vs Credit Score")
plt.show()

# 4. Box Plot
plt.figure(figsize=(6,4))
sns.boxplot(x="Loan_Status", y="Income", data=df)
plt.title("Income by Loan Status")
plt.show()

# 5. Important Patterns
print("\nImportant Patterns")
print("• Customers with higher income are mostly approved.")
print("• Higher credit scores increase loan approval chances.")
print("• Approved customers generally have higher income.")
print("• Loan approval mainly depends on income and credit score.")

# 6. Observations
print("\nObservations")
print("1. Customers with high income are more likely to get loans.")
print("2. Higher credit scores lead to higher approval rates.")
print("3. Rejected customers usually have lower income.")
print("4. Income and Credit Score have a positive relationship.")
print("5. Credit Score is an important factor in loan approval.")

# 7. Conclusion
print("\nConclusion")
print("The analysis shows that Income and Credit Score play a major
role in loan approval. Customers with higher values have a greater
chance of getting their loans approved.")

```

Banking Loan Approval Dataset

	Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type
Loan_Status					
0	C01	45000	720	200000	Salaried
Approved					
1	C02	25000	580	150000	Self-employed
Rejected					
2	C03	60000	750	300000	Salaried
Approved					
3	C04	22000	560	120000	Unemployed
Rejected					
4	C05	50000	710	250000	Salaried
Approved					
5	C06	30000	600	180000	Self-employed
Rejected					
6	C07	70000	780	350000	Salaried
Approved					
7	C08	28000	590	160000	Self-employed
Rejected					
8	C09	55000	730	270000	Salaried

Approved					
9	C10	24000	570	130000	Unemployed
Rejected					
10	C11	65000	760	320000	Salaried
Approved					
11	C12	35000	620	200000	Self-employed
Rejected					
12	C13	48000	700	240000	Salaried
Approved					
13	C14	26000	585	140000	Unemployed
Rejected					
14	C15	75000	790	400000	Salaried
Approved					
15	C16	32000	610	190000	Self-employed
Rejected					
16	C17	52000	725	260000	Salaried
Approved					
17	C18	27000	575	150000	Self-employed
Rejected					
18	C19	68000	770	330000	Salaried
Approved					
19	C20	23000	555	125000	Unemployed
Rejected					

Rows and Columns: (20, 6)

Data Types

Customer_ID	object
Income	int64
Credit_Score	int64
Loan_Amount	int64
Employment_Type	object
Loan_Status	object

dtype: object

Missing Values

Customer_ID	0
Income	0
Credit_Score	0
Loan_Amount	0
Employment_Type	0
Loan_Status	0

dtype: int64

Duplicate Values: 0

Mean

Income	43000.0
Credit_Score	664.0
Loan_Amount	223250.0

dtype: float64

Median

Income 40000.0

Credit_Score 660.0

Loan_Amount 200000.0

dtype: float64

Minimum

Income 22000

Credit_Score 555

Loan_Amount 120000

dtype: int64

Maximum

Income 75000

Credit_Score 790

Loan_Amount 400000

dtype: int64

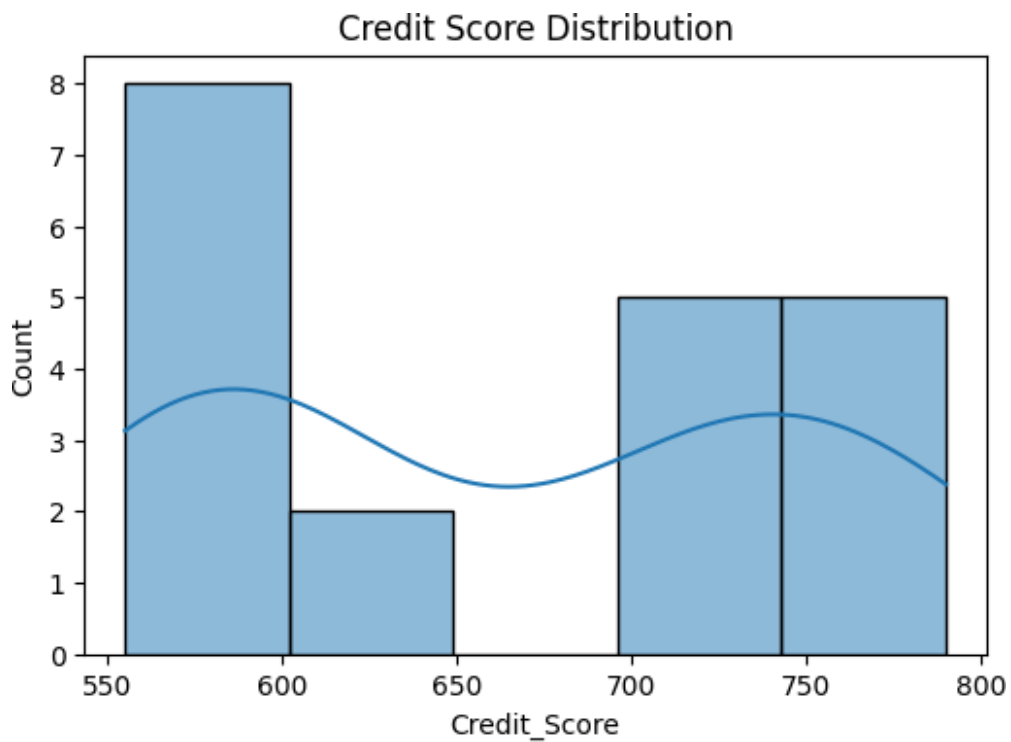
Standard Deviation

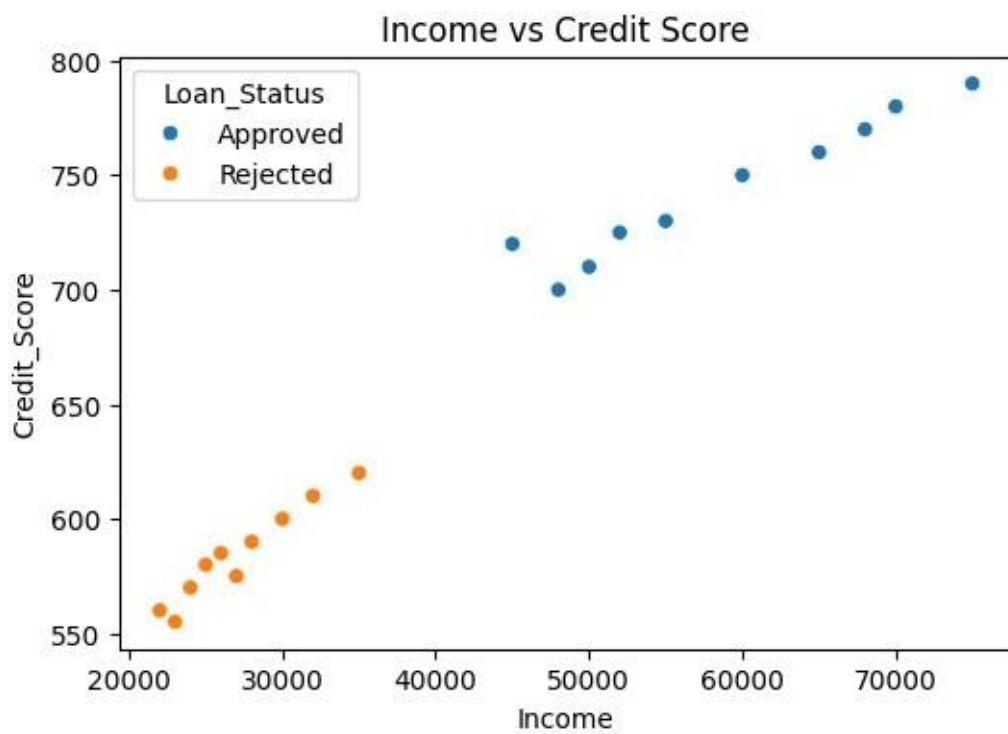
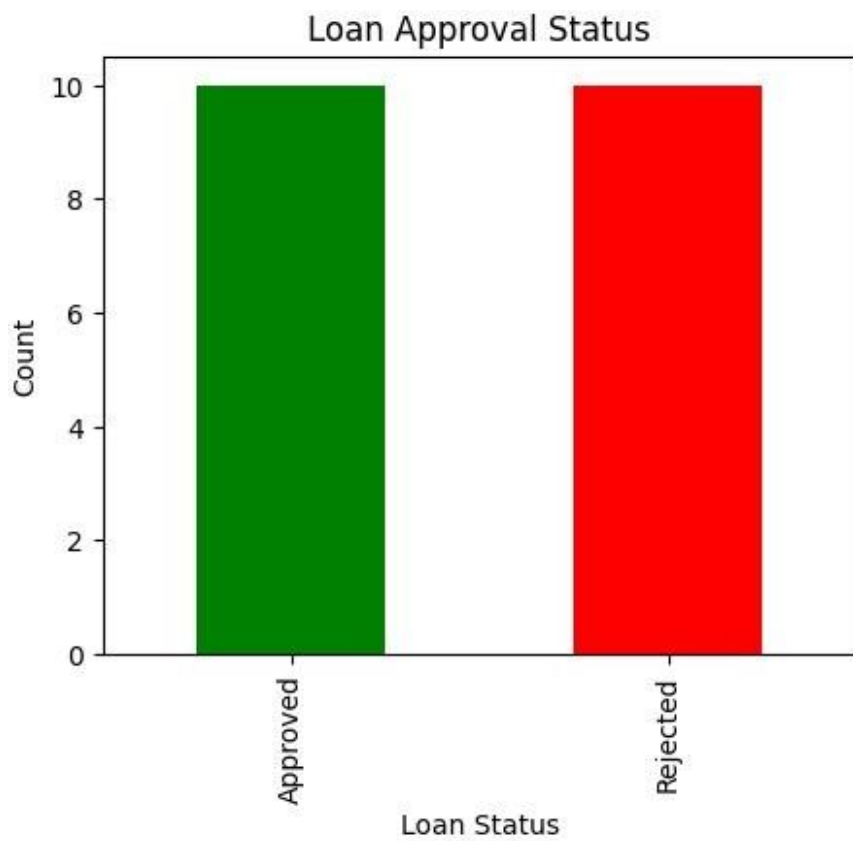
Income 17923.815383

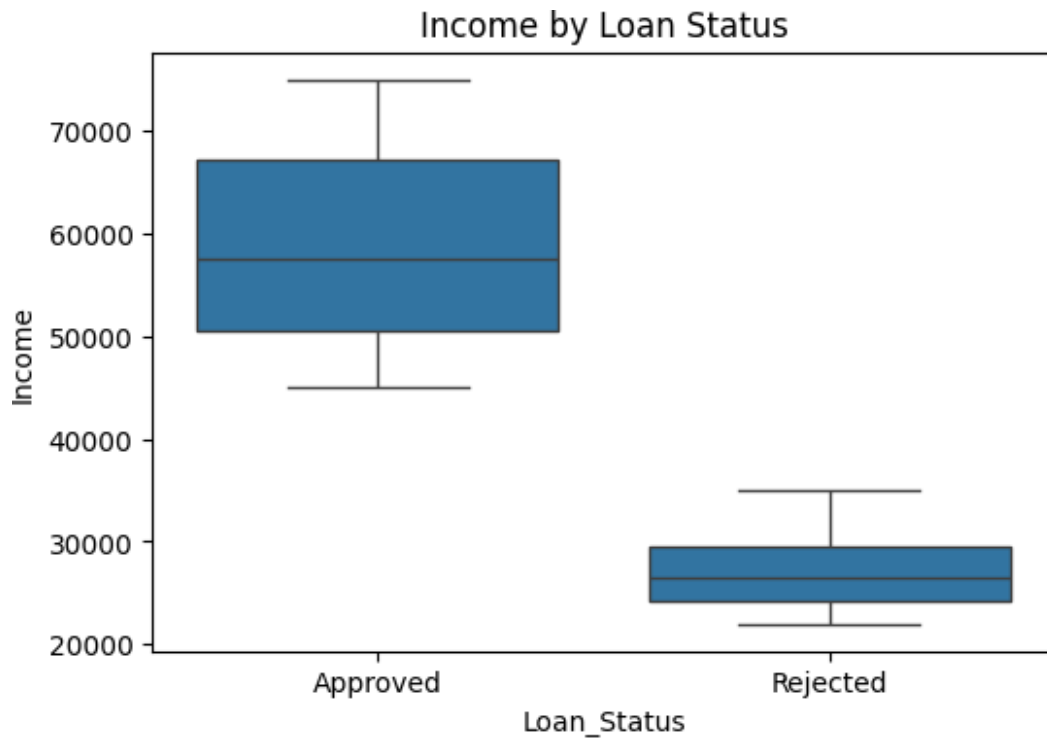
Credit_Score 85.526235

Loan_Amount 83733.207272

dtype: float64







Important Patterns

- Customers with higher income are mostly approved.
- Higher credit scores increase loan approval chances.
- Approved customers generally have higher income.
- Loan approval mainly depends on income and credit score.

Observations

1. Customers with high income are more likely to get loans.
2. Higher credit scores lead to higher approval rates.
3. Rejected customers usually have lower income.
4. Income and Credit Score have a positive relationship.
5. Credit Score is an important factor in loan approval.

Conclusion

The analysis shows that Income and Credit Score play a major role in loan approval. Customers with higher values have a greater chance of getting their loans approved.

```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
    "Farm_ID":
```

```

["F01", "F02", "F03", "F04", "F05", "F06", "F07", "F08", "F09", "F10",
"F11", "F12", "F13", "F14", "F15", "F16", "F17", "F18", "F19", "F20"],
    "Rainfall_mm": [850, 600, 900, 500, 780, 650, 920, 720, 550, 800,
                    880, 620, 760, 940, 580, 820, 700, 960, 540, 790],
    "Temperature": [28, 32, 27, 34, 29, 31, 26, 30, 33, 28,
                   27, 32, 29, 26, 33, 28, 30, 25, 34, 29],
    "Fertilizer_kg": [60, 45, 65, 40, 55, 48, 70, 52, 42, 58,
                    62, 46, 54, 72, 44, 60, 50, 75, 41, 56],
    "Soil_Type": ["Loamy", "Sandy", "Loamy", "Sandy", "Clay",
                  "Sandy", "Loamy", "Clay", "Sandy", "Loamy",
                  "Loamy", "Sandy", "Clay", "Loamy", "Sandy",
                  "Clay", "Clay", "Loamy", "Sandy", "Clay"],
    "Crop_Yield_kg": [3200, 2100, 3500, 1800, 3000, 2300, 3700, 2800, 1900, 3100,
                    3400, 2200, 2950, 3800, 2000, 3150, 2700, 3900, 1850, 3050]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display Dataset
print("Agriculture Crop Yield Dataset")
print(df)

# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

# 2. Data Types
print("\nData Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

```

```

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# ----- Charts -----

# 1. Histogram
plt.figure(figsize=(6,4))
sns.histplot(df["Crop_Yield_kg"], bins=5, kde=True)
plt.title("Crop Yield Distribution")
plt.xlabel("Crop Yield (kg)")
plt.ylabel("Frequency")
plt.show()

# 2. Bar Chart
soil_yield = df.groupby("Soil_Type")["Crop_Yield_kg"].mean()

plt.figure(figsize=(6,4))
soil_yield.plot(kind="bar", color="green")
plt.title("Average Crop Yield by Soil Type")
plt.xlabel("Soil Type")
plt.ylabel("Average Crop Yield (kg)")
plt.show()

# 3. Scatter Plot
plt.figure(figsize=(6,4))
sns.scatterplot(x="Rainfall_mm", y="Crop_Yield_kg", hue="Soil_Type",
data=df)
plt.title("Rainfall vs Crop Yield")
plt.show()

# 4. Box Plot
plt.figure(figsize=(6,4))
sns.boxplot(x="Soil_Type", y="Crop_Yield_kg", data=df)
plt.title("Crop Yield by Soil Type")
plt.show()

# 5. Important Patterns
print("\nImportant Patterns")
print("• Higher rainfall generally results in higher crop yield.")
print("• Loamy soil produces better crop yield than Sandy soil.")
print("• Higher fertilizer usage is associated with higher crop
production.")
print("• Temperature has a moderate impact on crop yield.")

# 6. Observations
print("\nObservations")
print("1. Farms with higher rainfall produce higher crop yield.")

```

```

print("2. Loamy soil has the highest average crop yield.")
print("3. Sandy soil gives comparatively lower crop yield.")
print("4. Fertilizer usage positively influences crop production.")
print("5. Rainfall and fertilizer are the major factors affecting crop yield.")

```

```

# 7. Conclusion

```

```

print("\nConclusion")

```

```

print("The analysis shows that rainfall, soil type, and fertilizer usage significantly influence crop yield. Loamy soil with adequate rainfall and fertilizer gives the best crop production.")

```

Agriculture Crop Yield Dataset

	Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type
Crop_Yield_kg					
0	F01	850	28	60	Loamy
3200					
1	F02	600	32	45	Sandy
2100					
2	F03	900	27	65	Loamy
3500					
3	F04	500	34	40	Sandy
1800					
4	F05	780	29	55	Clay
3000					
5	F06	650	31	48	Sandy
2300					
6	F07	920	26	70	Loamy
3700					
7	F08	720	30	52	Clay
2800					
8	F09	550	33	42	Sandy
1900					
9	F10	800	28	58	Loamy
3100					
10	F11	880	27	62	Loamy
3400					
11	F12	620	32	46	Sandy
2200					
12	F13	760	29	54	Clay
2950					
13	F14	940	26	72	Loamy
3800					
14	F15	580	33	44	Sandy
2000					
15	F16	820	28	60	Clay
3150					
16	F17	700	30	50	Clay
2700					
17	F18	960	25	75	Loamy

3900					
18	F19	540	34	41	Sandy
1850					
19	F20	790	29	56	Clay
3050					

Rows and Columns: (20, 6)

Data Types

Farm_ID	object
Rainfall_mm	int64
Temperature	int64
Fertilizer_kg	int64
Soil_Type	object
Crop_Yield_kg	int64

dtype: object

Missing Values

Farm_ID	0
Rainfall_mm	0
Temperature	0
Fertilizer_kg	0
Soil_Type	0
Crop_Yield_kg	0

dtype: int64

Duplicate Values: 0

Mean

Rainfall_mm	743.00
Temperature	29.55
Fertilizer_kg	54.75
Crop_Yield_kg	2820.00

dtype: float64

Median

Rainfall_mm	770.0
Temperature	29.0
Fertilizer_kg	54.5
Crop_Yield_kg	2975.0

dtype: float64

Minimum

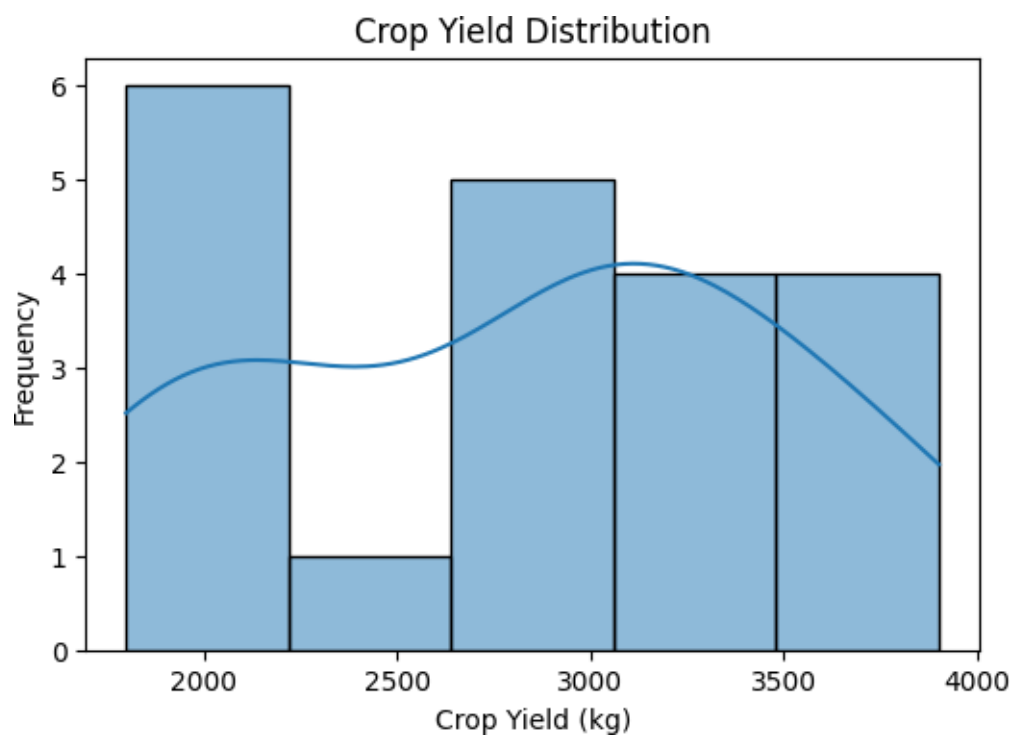
Rainfall_mm	500
Temperature	25
Fertilizer_kg	40
Crop_Yield_kg	1800

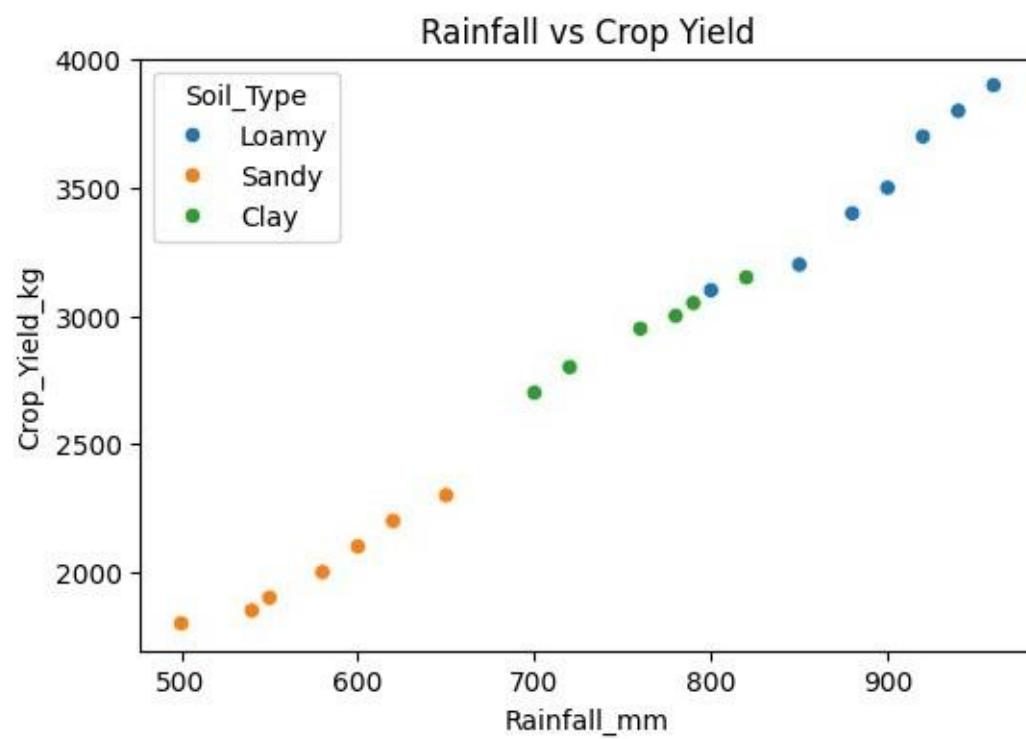
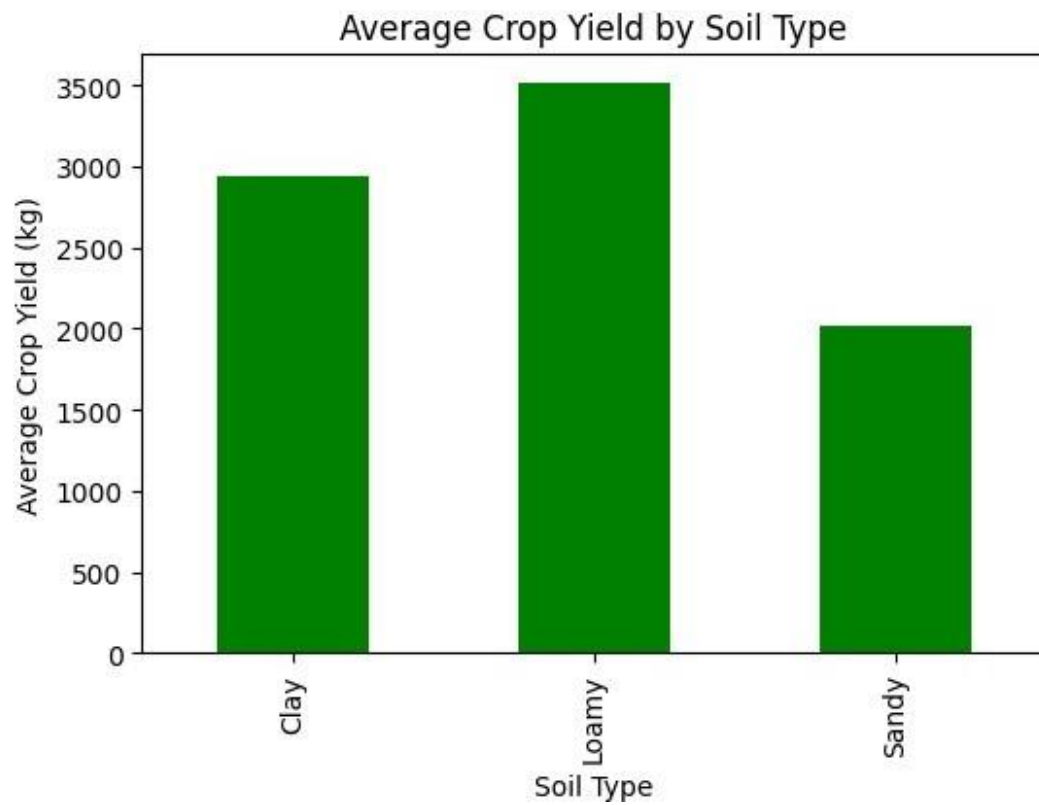
dtype: int64

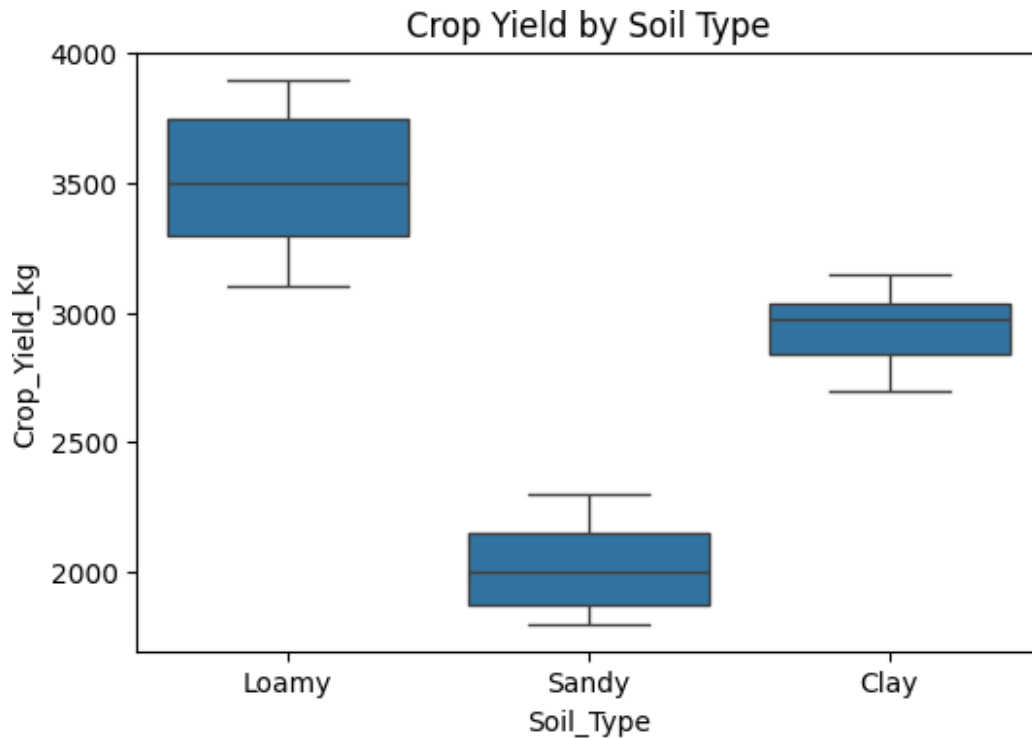
Maximum


```
Rainfall_mm      960
Temperature       34
Fertilizer_kg     75
Crop_Yield_kg    3900
dtype: int64
```

```
Standard Deviation
Rainfall_mm      144.335136
Temperature       2.762055
Fertilizer_kg    10.507516
Crop_Yield_kg    681.407213
dtype: float64
```







Important Patterns

- Higher rainfall generally results in higher crop yield.
- Loamy soil produces better crop yield than Sandy soil.
- Higher fertilizer usage is associated with higher crop production.
- Temperature has a moderate impact on crop yield.

Observations

1. Farms with higher rainfall produce higher crop yield.
2. Loamy soil has the highest average crop yield.
3. Sandy soil gives comparatively lower crop yield.
4. Fertilizer usage positively influences crop production.
5. Rainfall and fertilizer are the major factors affecting crop yield.

Conclusion

The analysis shows that rainfall, soil type, and fertilizer usage significantly influence crop yield. Loamy soil with adequate rainfall and fertilizer gives the best crop production.